

■ Patient Readmission Risk Analysis Report

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1. Project Overview

This report presents an end-to-end analysis of diabetic patient readmission risk using real hospital data from 130 US hospitals. The goal is to predict whether a patient will be readmitted within 30 days of discharge using machine learning.

2. Dataset Summary

Metric	Value
Total Patients	101,766
Total Features	43
Readmitted Patients	11,357
Not Readmitted	90,409
Readmission Rate	11.16%
Average Hospital Stay	4.40 days
Average Medications	16.02

3. Key Findings

- Patients with more inpatient visits are most likely to be readmitted
- Readmitted patients take significantly more medications (p-value < 0.05)
- Average hospital stay is 4.40 days across all patients
- Female patients slightly outnumber male patients in this dataset
- Only 11.16% of patients were readmitted within 30 days

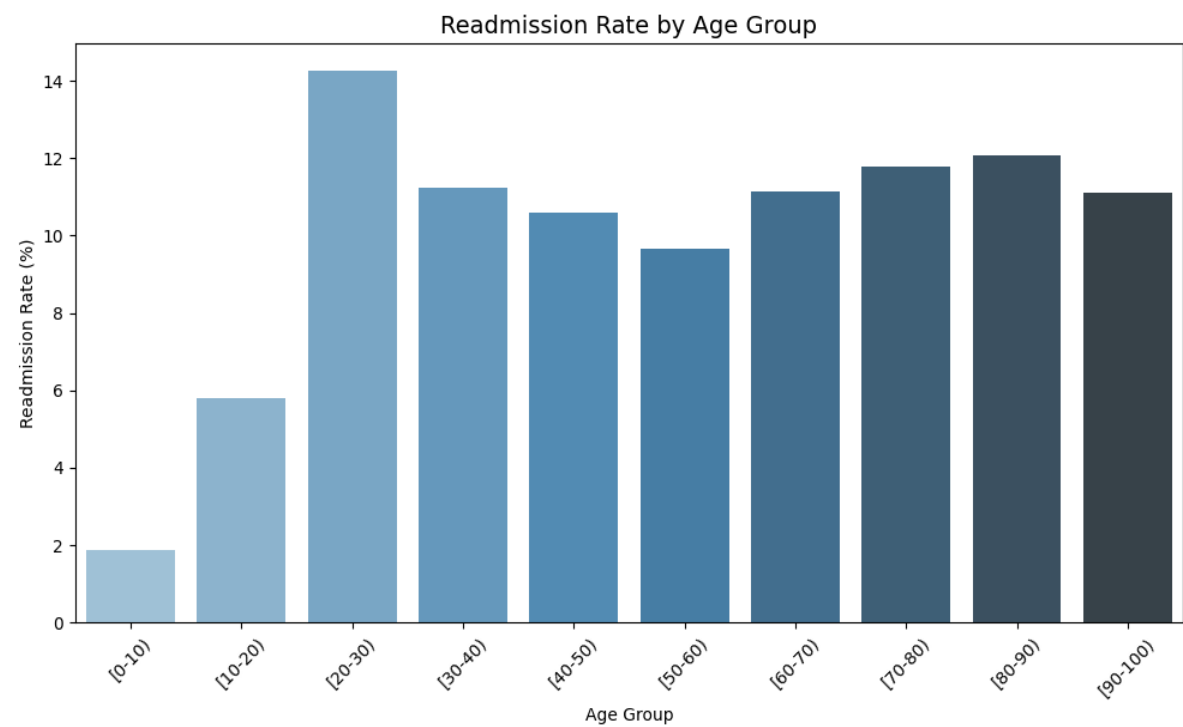
4. Model Performance

Metric	Value
Model Used	Logistic Regression
Training Patients	81,412

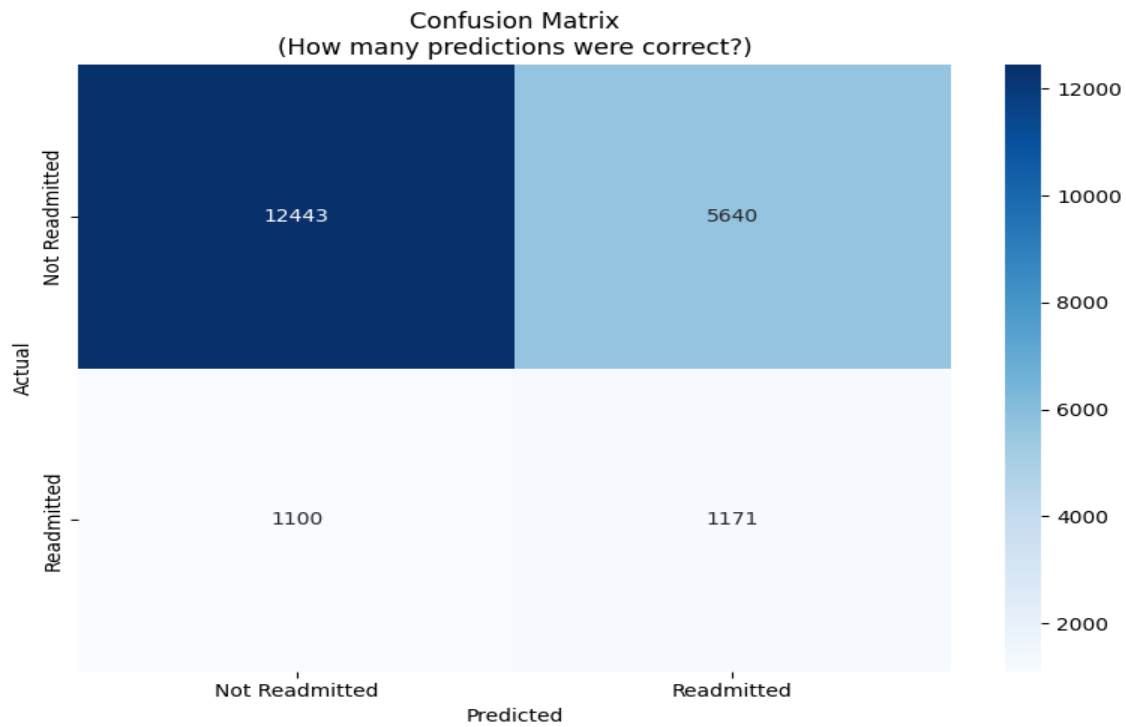
Testing Patients	20,354
Overall Accuracy	66.89%
Recall (Readmitted)	52%

5. Visualizations

Readmission Rate by Age Group



Model Confusion Matrix



Medication Distribution

